

HARSHAL KOLHE

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EDUCATION

Visvesvaraya National Institute of Technology, India

Bachelor of Technology in Chemical Engineering, Minor in Internet of Things

Aug' 23 – May' 27

CGPA: [8.66/10.0](#)

EXPERIENCE

Indian Institute of Technology, Roorkee

May' 25 – Jul' 25

Summer Research Intern | [\[Certificate\]](#)

- Developed a reinforcement learning-based control system for a quadcopter carrying a suspended payload, with active suppression of payload-induced oscillations during flight.
- Built a high-frequency data collection and processing pipeline that synchronized multi-source flight and payload unit data streams for reinforcement learning policy validation.
- Built a real-time autonomy stack that streamed 20+ state variables from CubeOrange at 100 Hz via MAVLink, performed onboard RL inference, and generated flight control commands for autonomous stabilization.

PROJECTS

AutoGrade — Intelligent Examination Evaluation & Verification System | [\[GitHub\]](#)

May' 26 – Jul' 26

- Built a handwritten OCR system to convert scanned student answer sheets into machine-readable text for automated evaluation.
- Developed an OCR pipeline using Pixtral Large, PaddleOCR, and EasyOCR with image preprocessing, multi-page PDF support, and batch processing.
- Processed 4,274 student answers and integrated extracted text with question segmentation and semantic scoring for automated evaluation.

Dual UAV Scan and Spray Architecture for Precision Agriculture | [\[GitHub\]](#)

Jan' 26 – Apr ' 26

- Developed a real-time onboard crop disease detection pipeline using OpenCV and HSV-based image segmentation, under varying outdoor lighting conditions through dual-mask filtering and ND-filter-assisted preprocessing.
- Built a dual-nozzle spraying system with real-time nozzle diameter selection based on detected plant clusters; developed the Raspberry Pi-based control algorithm and designed, fabricated, and assembled the control PCB for automated operation.
- Designed and fabricated a custom carbon-fiber quadcopter frame capable of carrying a 10 kg fluid payload.

Multiagent Hybrid Disaster Rescue Drones | [\[GitHub\]](#)

Jul' 25 – Jan' 26

- Developed a two-drone system that searches in a lawnmower pattern, detects humans, and diverts to deliver relief packages.
- Fine-tuned and deployed a YOLO object detection model for aerial imagery, achieving 90% human detection accuracy at 8m, and integrated it into an end-to-end autonomous localisation pipeline that generated target GPS coordinates and performed dynamic UAV redirection with cluster-based filtering.
- Developed the core telemetry processing layer for a multi-UAV visualisation platform, combining MAVLink data streams and enabling real-time tracking, status monitoring, and mission visualisation.

PUBLICATIONS

- S. Kshirsagar, **H. Kolhe**, R. Deshmukh, K. Ayush, S. Dhalwar, S. Biswas, M. Limaye, and S. S. Chiddarwar, "A Multi-Agent Drone System for Real-Time Victim Detection and Aid Delivery," presented at the International Conference on Advances in Mechanical and Manufacturing Systems (ICAMMS), VNIT Nagpur, India, 2026.
- A. Sharma, M. Ravikiran, S. Chakrabarty, R. Saluja, M. Limaye, **H. Kolhe**, and S. Samiron, "Prompted to Fly: Translating Free-Form Instructions into Schema-Constrained Mission Generation for UAVs using LLMs," *2025 IEEE International Conference on Big Data (BigData)*, Macau, China, 2025, pp. 7278-7286, doi: [10.1109/BigData66926.2025.11401432](https://doi.org/10.1109/BigData66926.2025.11401432).

SKILLS

Languages: C, C++, Python

Softwares and Tools: ArduPilot, QGroundControl, Mission Planner, Linux, ROS2

Libraries: OpenCV, Ultralytics (YOLOv8), NumPy, Pandas, Matplotlib, DroneKit, MAVSDK, Mavlink

EXTRACURRICULAR ACTIVITIES

- Secured the [Best Paper Presentation Award](#) at the ICAMMS Conference for presenting research work on autonomous systems.
- Secured the Ignite Prize in **NIDAR**, a national-level UAV autonomy Competition, placing among the top 5 out of 100+ teams.
- Secured [First Runner-up](#) prize at the Drone Helix Competition, SVNIT Surat.
- Vice Chairman of IvLabs, Robotics and Artificial Intelligence Lab at VNIT, Nagpur.